

CHAPTER 1:

Java Fundamental

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Programming Paradigm

- Linear Programming
- Procedural Programming
- Structured Programming
- Object Oriented Programming
- Functional Programming
- Logic Programming



Class Activity

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Linear Programming

- **linear programming** (LP) is a technique for optimization
- Linear programming can be applied to various fields of study.
- Most extensively it is used in business and economic situations, but can also be utilized for some engineering problems.
- Some industries that use linear programming models include transportation, energy, telecommunications, and manufacturing.
- It has proved useful in modeling diverse types of problems in planning, routing, scheduling, assignment, and design.

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Procedural Programming

- Procedural programming can be describe as specifying the steps the program must take to reach the desired state
- Based upon the concept of the *procedure call*.
- Procedures, also known as routines, subroutines, methods, or functions
- Advantages
  - The ability to re-use the same code at different places in the program without duplication
  - The ability to be strongly modular or structured.

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Structured Programming

- It states that three ways of combining programs—sequencing, selection, and iteration
  - Sequence" refers to an ordered execution of statements.
  - In "selection" one of a number of statements is executed depending on the state of the program. This is usually expressed with keywords such as if..then..else..endif, switch, or case.
  - In "repetition" a statement is executed until the program reaches a certain state or operations are applied to every element of a collection. This is usually expressed with keywords such as while, repeat, for or do..until.

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Object Oriented Programming

- Object-oriented programming (OOP) is a programming paradigm that uses "objects" – data structures consisting of datafields and methods together with their interactions – to design applications and computer programs.

- Programming techniques may include features such as information hiding, data abstraction, encapsulation, modularity, polymorphism, and inheritance.

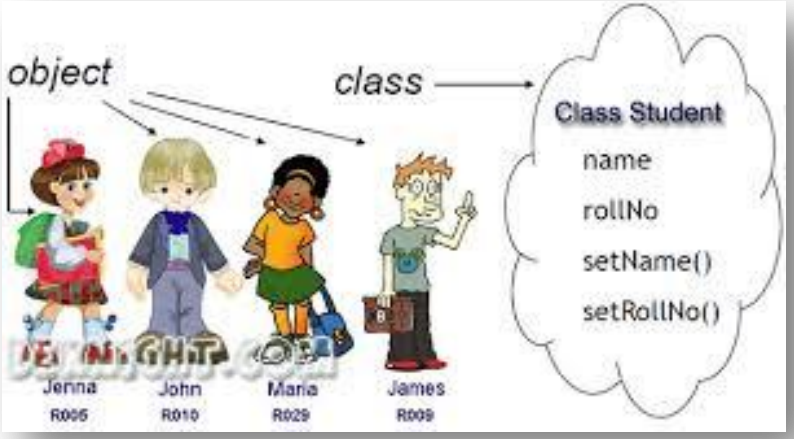
- It was not commonly used in mainstream software application development until the early 1990s.

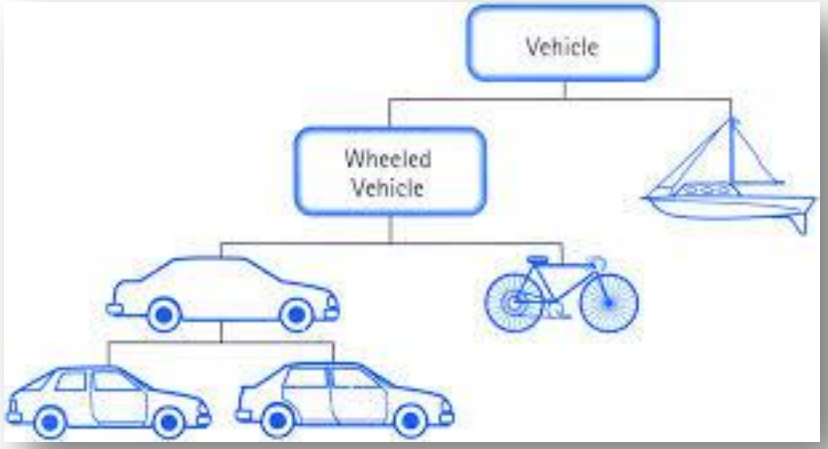
- Many modern programming languages now support OOP.

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OOP Concepts

Method of implementation in which programs are organized as cooperative collection of objects, each of which represents an instance of a class, and whose classes are all members of a hierarchy of classes united via inheritance relationships.





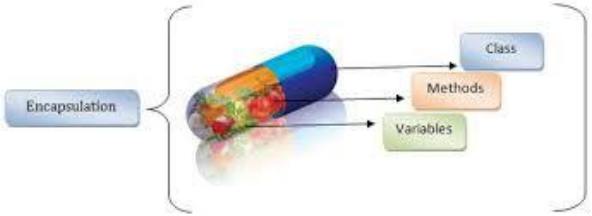
OOP Concepts

4 Principles

Abstraction: Abstraction denotes the essential characteristics of an object that distinguish it from all other kinds of objects and thus provide crisply defined conceptual boundaries, relative to the perspective of the viewer.



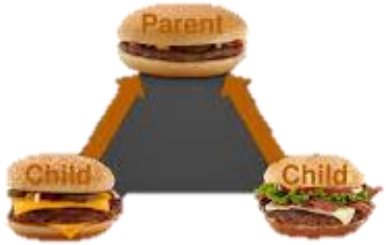
Encapsulation: process of compartmentalizing the elements of an abstraction that constitute its structure and behavior ; encapsulation serves to separate the contractual interface of an abstraction and its implementation.



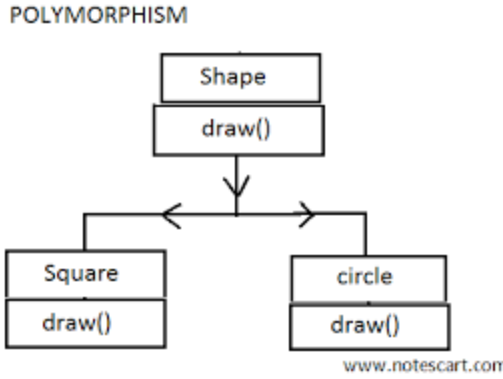
OOP Concepts

4 Principles

Inheritance: the process by which one object acquires the properties of another object. A relationship among classes, wherein one class shares the structure and behavior defined in 1 or more other classes.



Polymorphism: the existence of the classes or methods in different forms or single name denoting different implementations.



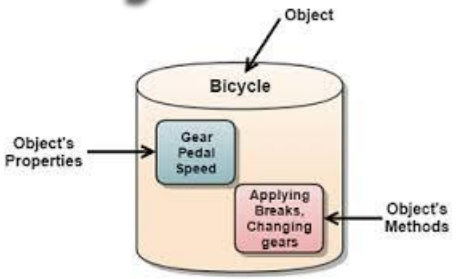
OOP Concepts

Class



A set of objects that share a common structure and behavior. Blueprint for creating different objects which defines its properties and behaviors. A class can contain fields and methods to describe the state, behavior and identity of an object.

Object



Instance of a particular class. An object exhibits the properties (structures) and behaviors defined by its class.

Method

Members of a class that provide a service for an object or perform task and return information when task completed.
- Statements within the methods determine what program does.

Java History

- Java is a programming language originally developed by James Gosling at Sun Microsystems and released in 1995
- Language derives much of its syntax from C and C++
- Has a simpler object model and fewer low-level facilities.



Duke, the Java mascot



Paradigm	Object-oriented, structured, imperative
Appeared in	1995
Designed by	Sun Microsystems
Stable release	Java Standard Edition 6 (1.6.0_16)
Typing discipline	Static, strong, safe, nominative, manifest
Major implementations	Numerous

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Introduction

What is JAVA?

Combination of 3 elements:



JAVA

Java Programming Language

High-level object-oriented programming language

How It Works?

✓All source code is first written in plain text files ending with the .java extension.

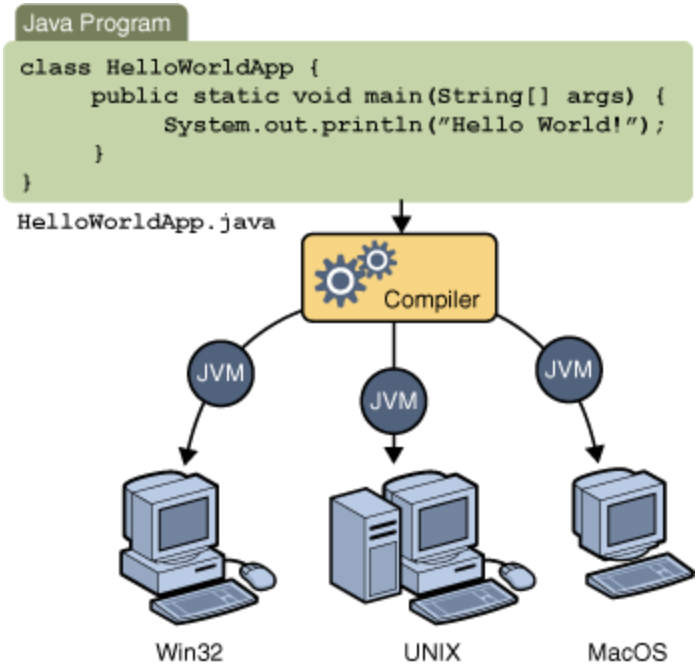
✓Those source files are then compiled into .class files by the javac compiler.

✓A .class file does not contain code that is native to your processor; it instead contains *bytecodes* — the machine language of the Java Virtual Machine (Java VM).

✓The java launcher tool then runs your application with an instance of the Java Virtual Machine.

Java Virtual Machine





JVM: virtual machine that interpret the bytecode for a computer's processor so that it can perform a Java program's intructions.

Java Virtual Machine

- Java was designed to allow application programs to be built that could be run on any platform without having to be rewritten or recompiled by the programmer for each separate platform.

- Because the Java VM is available on many different operating systems, the same .class files are capable of running on multiple platforms, Microsoft Windows, the Solaris ™ Operating System (Solaris OS), Linux, or Mac OS.

Java Platform

A *platform* is the hardware or software environment in which a program runs.

The Java platform differs from most other platforms in that it's a software-only platform that runs on top of other hardware-based platforms.

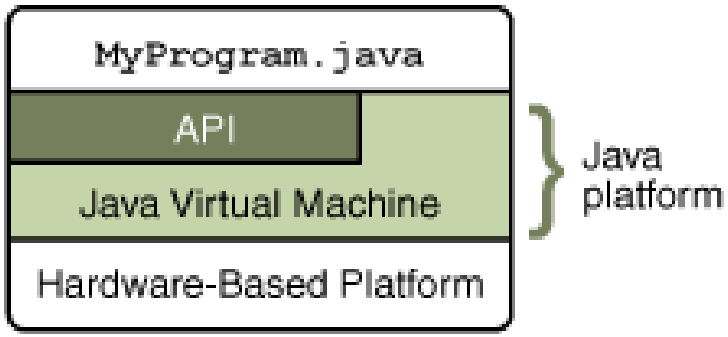
The Java platform (JRE) has two components:

- The *Java Virtual Machine*
- The *Java Application Programming Interface* (API)

JAVA PLATFORM

- **Java Virtual Machine**: the base for the Java platform and is ported onto various hardware-based platforms.

- **The API**: a large collection of ready-made software components that provide many useful capabilities. It is grouped into libraries of related classes and interfaces; these libraries are known as *packages*.



Java Software

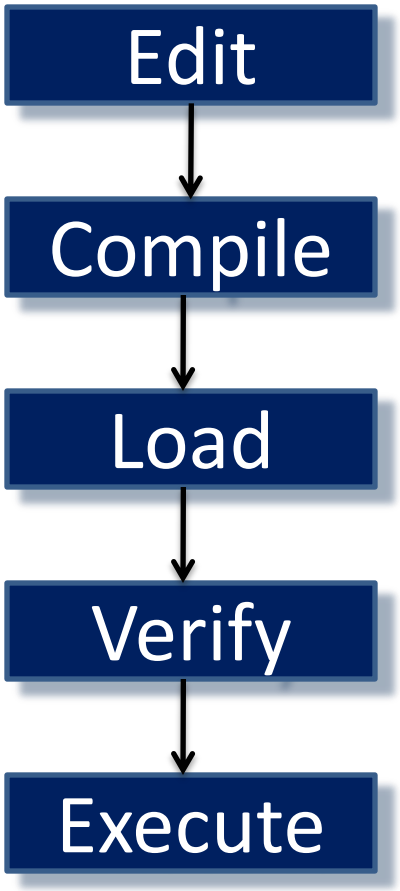
Java development tools:

For compiling, running, monitoring, debugging, and documenting your applications.
NetBeans, JDK, JCreator, Blue JCreator, Eclipse, etc.

User Interface Toolkits:

To create sophisticated Graphical User Interfaces (GUIs).
Swing and Java 2D toolkits

Java Environment



Source code.
File .java produced

Translate program to bytecodes
File .class produced

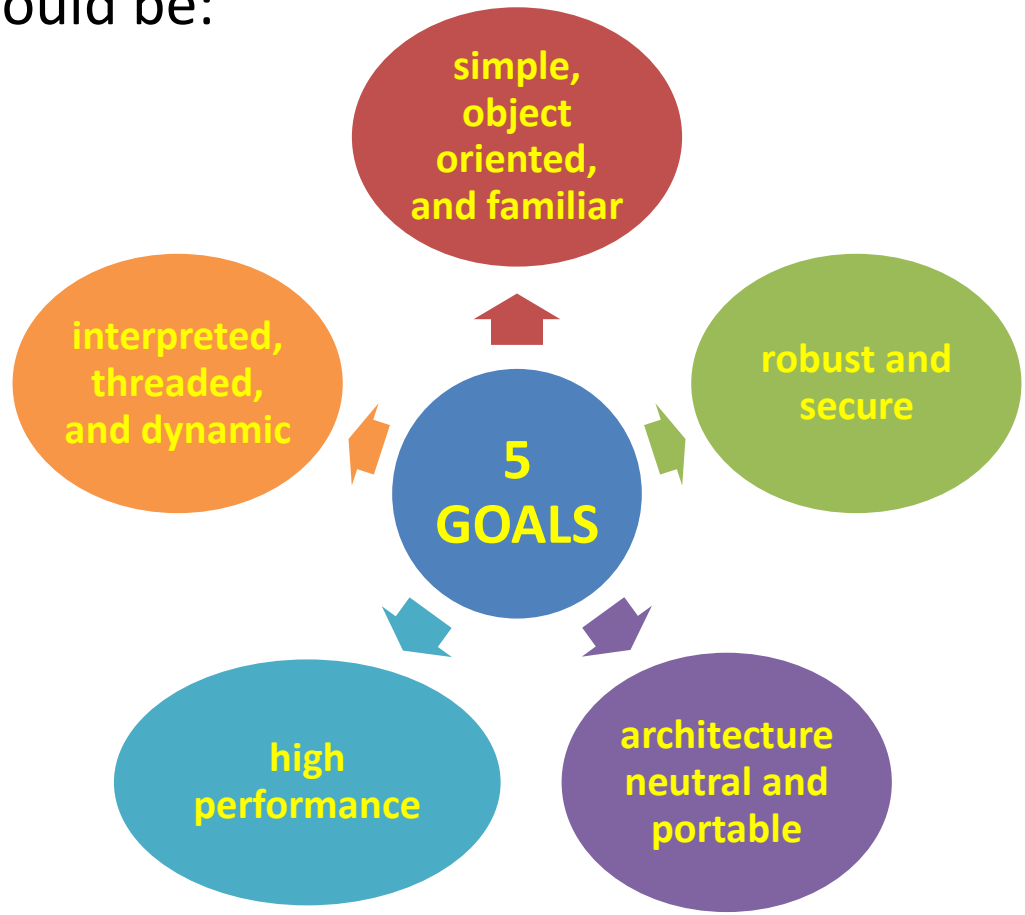
Program load in memory before execution

Bytecodes verifier confirms all codes are valid – strong security

Bytecodes are interpreted one at a time – perform actions.

Java Programming Principles (1)

There were **five primary goals** in the creation of the Java language: It should be:



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Java Programming Principles (1)

There were five primary goals in the creation of the Java language:

- It should be "simple, object oriented, and familiar".
- It should be "robust and secure“.
- It should be "architecture neutral and portable".
- It should execute with "high performance".
- It should be "interpreted, threaded, and dynamic".

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Java Programming Principles (2)

Advantages:

- Simple :- no pointer arithmetic, no structure and union, no pre-processor.
- Object-Oriented :- everything except simple types are objects, even the program itself is organized into class.
- Both compiled and interpreted :- Java code must be compiled into byte-codes (machine-independent) and run on Java Virtual Machine JVM (eg Browser), multi-platform.
- Multi-Threaded :- allows simultaneous activities in a program.
- Automatic Garbage-Collected :- programmers do not need to care about the de-allocation of claimed memory.
- Secure :- e.g. direct access of memory address is prohibited.

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Java Technology Editions

Java Platform, Standard Edition (Java SE)

For desktop PC, workstations and powerful consumer devices capable of sophisticated graphical interface environment, usually networked

Java Platform, Enterprise Edition (Java EE)

For creating large and powerful server-side and distributed applications, running on certain application servers like Weblogic, Websphere, jboss.. Etc.

Achieving fat-server, thin-client network (not necessarily web) application

Java Platform, Micro Edition (Java ME)

Devices includes PDA, set top box, TV box, and Java-enabled Mobile phones…etc

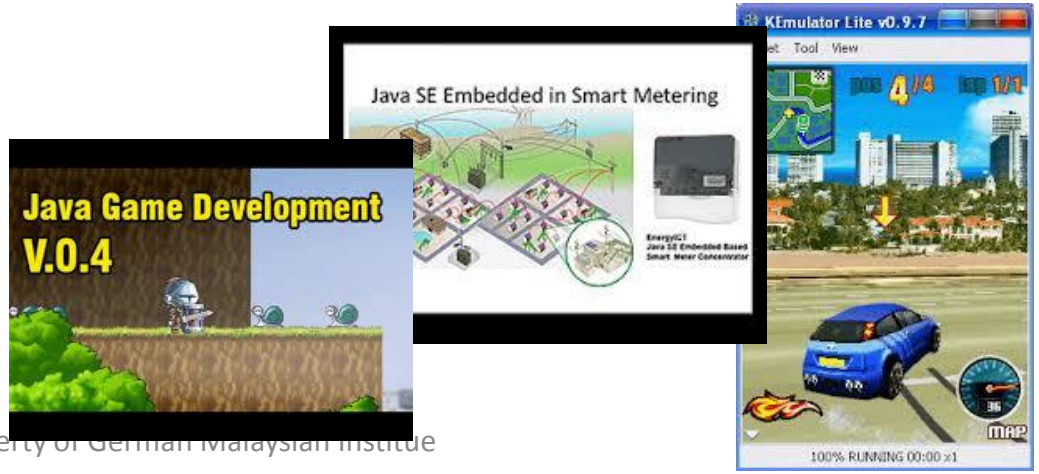
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Java Applications

- ✓Web based applications
- ✓Financial applications
- ✓Gaming applications
- ✓Embedded systems
- ✓Distributed enterprise applications
- ✓Mobile applications
- ✓Image processors
- ✓Desktop applications

and many more.





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Netbeans

- **NetBeans** began in 1996 as Xelfi, a Java IDE student project under the guidance of the Faculty of Mathematics and Physics at Charles University in Prague.
- In 1997 Roman Staněk formed a company around the project and produced commercial versions of the NetBeans IDE until it was bought by Sun Microsystems in 1999.
- Sun open-sourced the NetBeans IDE in June of the following year. The NetBeans community has since continued to grow.

- **NetBeans** refers to both a platform for the development of applications for the network using Java, JavaScript, PHP, Python, Ruby, Groovy, C, and C++, and an integrated development environment (IDE) developed using the NetBeans Platform

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Netbeans



Screenshot [show]

Developer(s)	Sun Microsystems
Stable release	6.7.1 / 2009-07-27; 3 months ago
Written in	Java
Operating system	Multi-platform
Available in	Multilingual
Type	Java IDE
License	CDDL or GPL2 + "certain source files" allow classpath exception
Website	www.netbeans.org

Hello World!!

```
public class HelloWorld {
     public static void main(String[] args) {
          System.out.println("Hello World!!"); }
     //End of main }
//End of HelloWorld Class
```

Output:
Hello World!!

Printing Method in Netbean

- System.out.printl(“Hello world)”;
- System.out.print(“I am Poporo”);
- System.out.format(“My height is %.2f cm”, value1);

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